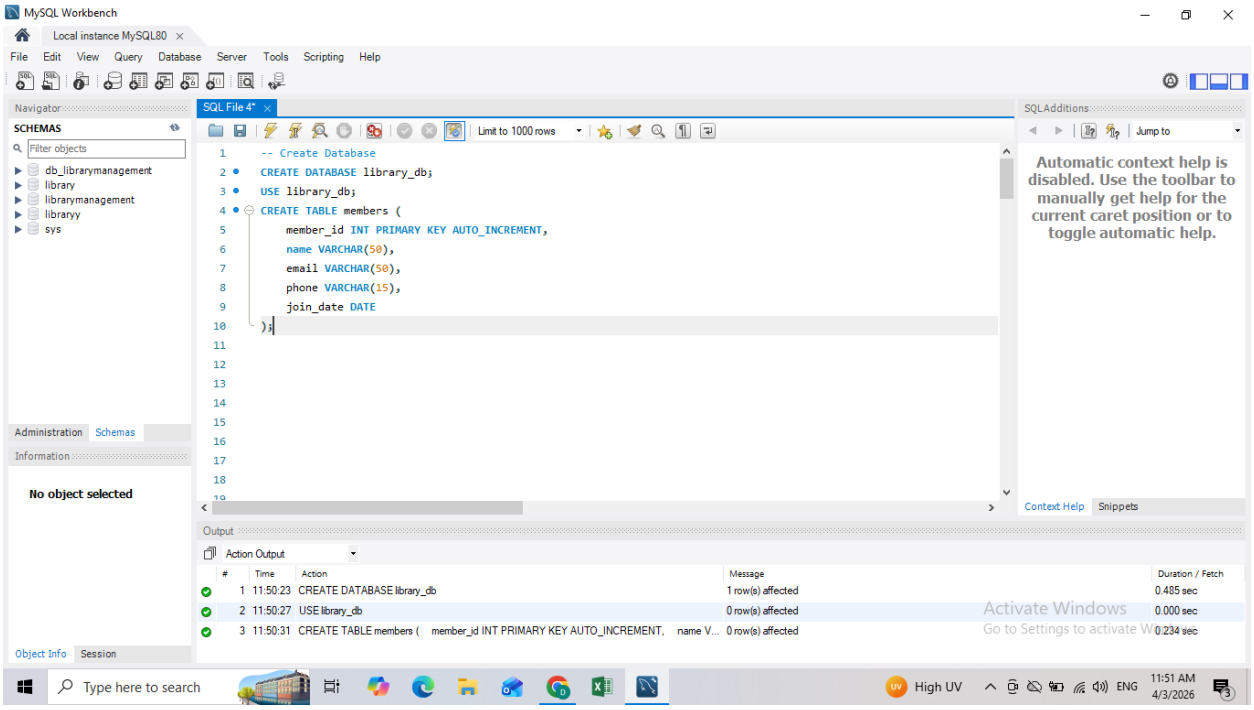
CREATING DATABASE, TABLES FOR MEMBERS, BOOKS, ISSUED BOOKS



MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 4*

```sql
-- Create Database
CREATE DATABASE library_db;
USE library_db;
CREATE TABLE members (
    member_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    email VARCHAR(50),
    phone VARCHAR(15),
    join_date DATE
);
CREATE TABLE books (
    book_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(100),
    author VARCHAR(50),
    category VARCHAR(50),
    available_copies INT
);
```

Output

#	Time	Action	Message	Duration / Fetch
1	11:50:23	CREATE DATABASE library_db	1 row(s) affected	0.485 sec
2	11:50:27	USE library_db	0 row(s) affected	0.000 sec
3	11:50:31	CREATE TABLE members (member_id INT PRIMARY KEY AUTO_INCREMENT, name V...	0 row(s) affected	0.234 sec
4	11:52:42	CREATE TABLE books (book_id INT PRIMARY KEY AUTO_INCREMENT, title VARCH...	0 row(s) affected	0.093 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 4*

```sql
);
CREATE TABLE books (
    book_id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(100),
    author VARCHAR(50),
    category VARCHAR(50),
    available_copies INT
);
CREATE TABLE issued_books (
    issue_id INT PRIMARY KEY AUTO_INCREMENT,
    member_id INT,
    book_id INT,
    issue_date DATE,
    return_date DATE,

    FOREIGN KEY (member_id) REFERENCES members(member_id),
    FOREIGN KEY (book_id) REFERENCES books(book_id)
);
```

Output

#	Time	Action	Message	Duration / Fetch
2	11:50:27	USE library_db	0 row(s) affected	0.000 sec
3	11:50:31	CREATE TABLE members (member_id INT PRIMARY KEY AUTO_INCREMENT, name...	0 row(s) affected	0.234 sec
4	11:52:42	CREATE TABLE books (book_id INT PRIMARY KEY AUTO_INCREMENT, title VARC...	0 row(s) affected	0.093 sec
5	11:54:34	CREATE TABLE issued_books (issue_id INT PRIMARY KEY AUTO_INCREMENT, me...	0 row(s) affected	0.313 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 4*

```
23        return_date DATE,
24
25        FOREIGN KEY (member_id) REFERENCES members(member_id),
26        FOREIGN KEY (book_id) REFERENCES books(book_id)
27    );
28    desc members;
```

Result Grid

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
name	varchar(50)	YES		NULL	
email	varchar(50)	YES		NULL	
phone	varchar(15)	YES		NULL	
join_date	date	YES		NULL	

Result 1

Output

#	Time	Action	Message	Duration / Fetch
3	11:50:31	CREATE TABLE members (member_id INT PRIMARY KEY AUTO_INCREMENT, nam...	0 row(s) affected	0.234 sec
4	11:52:42	CREATE TABLE books (book_id INT PRIMARY KEY AUTO_INCREMENT, title VARC...	0 row(s) affected	0.093 sec
5	11:54:34	CREATE TABLE issued_books (issue_id INT PRIMARY KEY AUTO_INCREMENT, m...	0 row(s) affected	0.313 sec
6	11:55:13	desc members	5 row(s) returned	0.125 sec / 0.000 sec

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

SQL File 4*

```
24
25        FOREIGN KEY (member_id) REFERENCES members(member_id),
26        FOREIGN KEY (book_id) REFERENCES books(book_id)
27    );
28    desc members;
29    desc books;
```

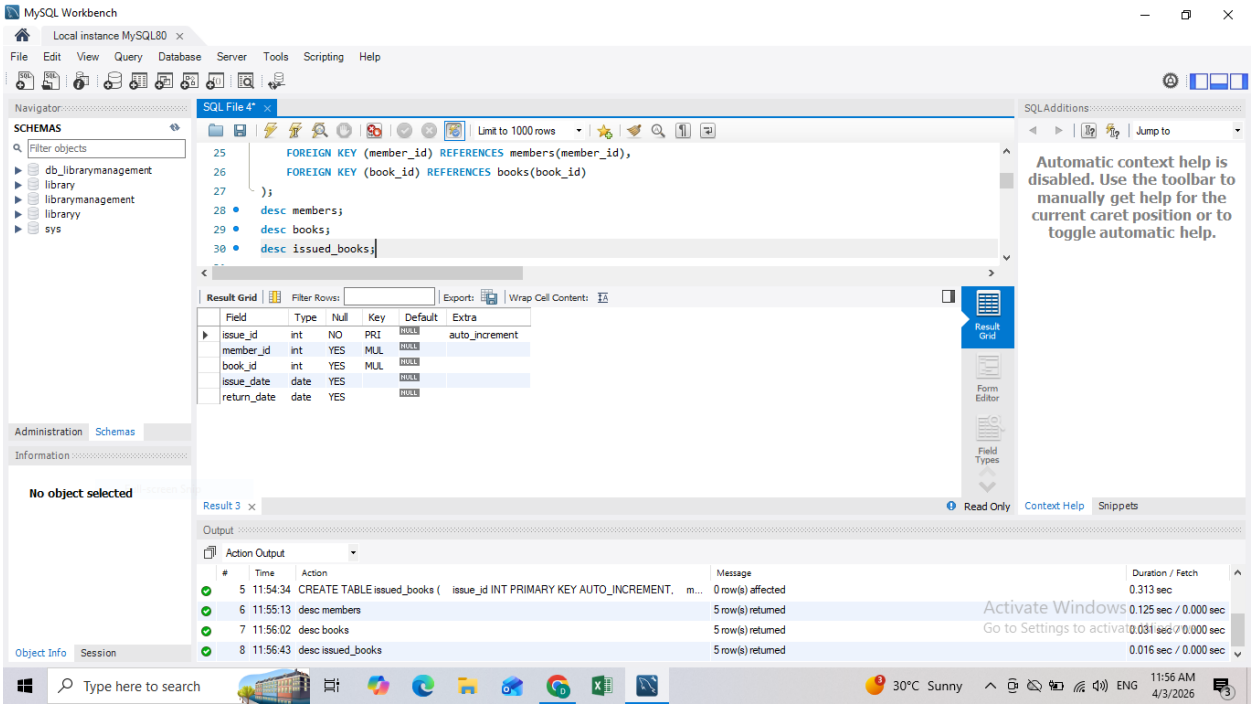
Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(100)	YES		NULL	
author	varchar(50)	YES		NULL	
category	varchar(50)	YES		NULL	
available_copies	int	YES		NULL	

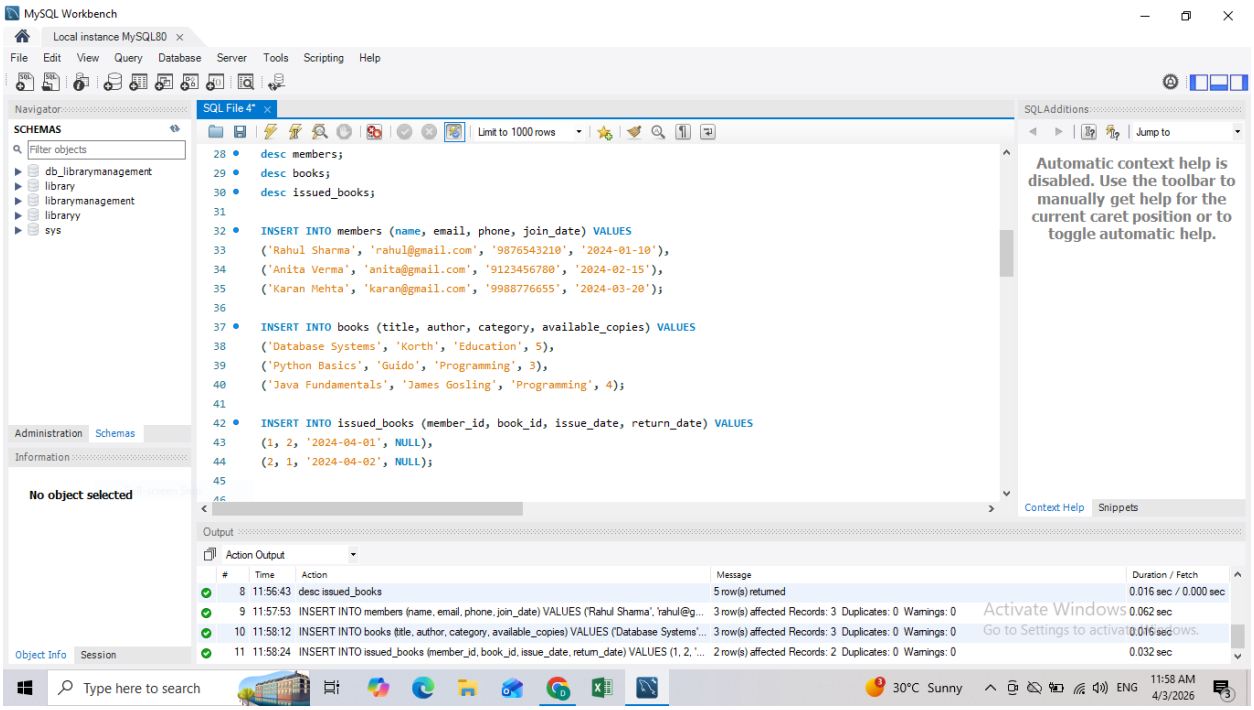
Result 2

Output

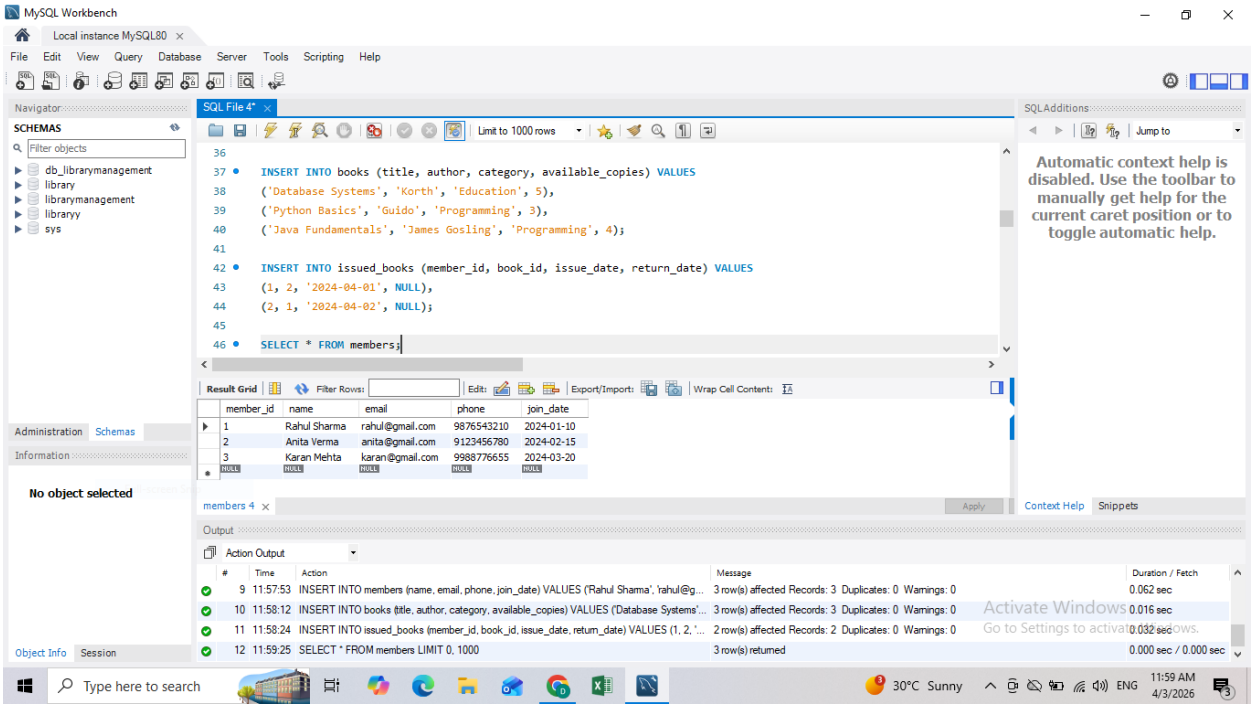
#	Time	Action	Message	Duration / Fetch
4	11:52:42	CREATE TABLE books (book_id INT PRIMARY KEY AUTO_INCREMENT, title VARC...	0 row(s) affected	0.093 sec
5	11:54:34	CREATE TABLE issued_books (issue_id INT PRIMARY KEY AUTO_INCREMENT, m...	0 row(s) affected	0.313 sec
6	11:55:13	desc members	5 row(s) returned	0.125 sec / 0.000 sec
7	11:56:02	desc books	5 row(s) returned	0.031 sec / 0.000 sec

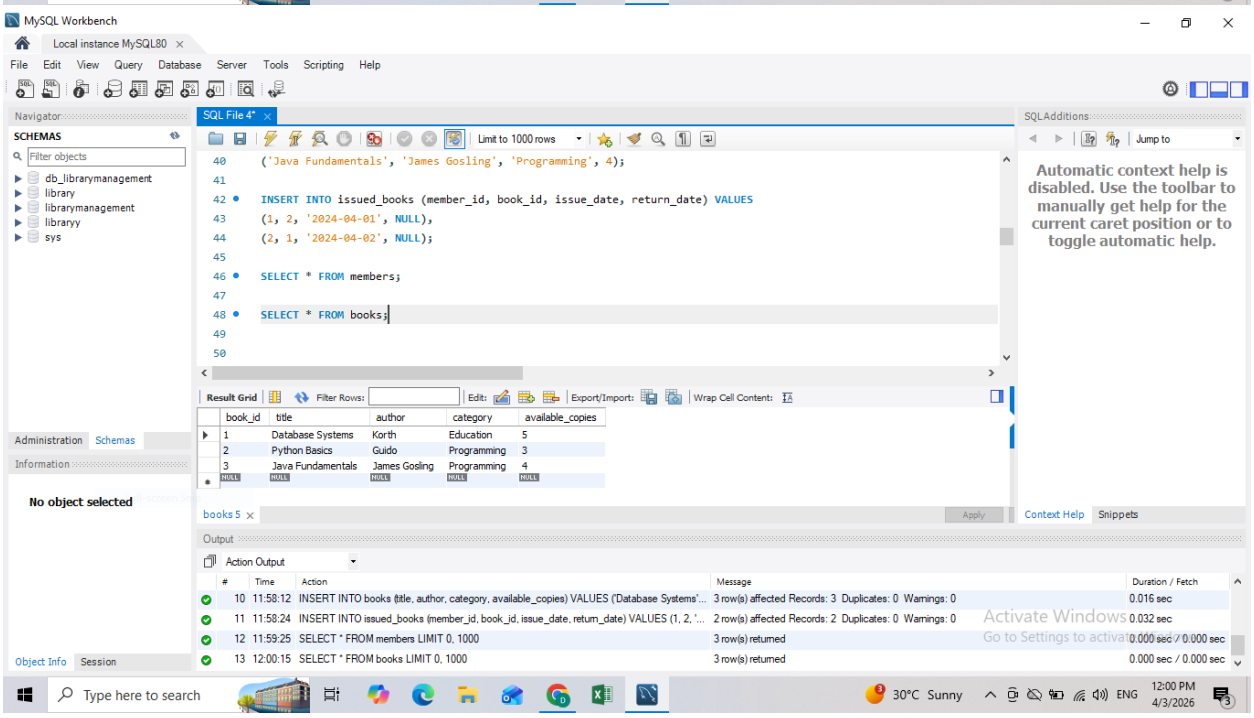


INSERTING DATA INTO MEMBERS, BOOKS, ISSUED BOOKS

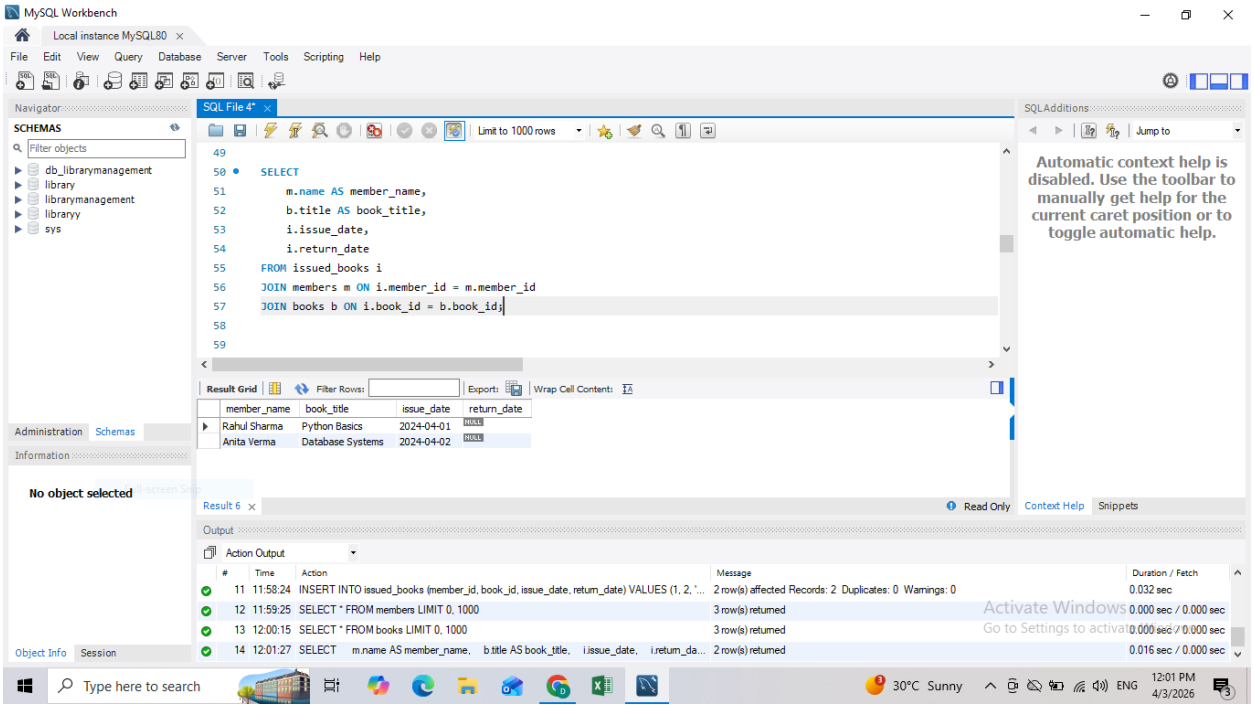


DISPLAYING MEMBERS, BOOKS

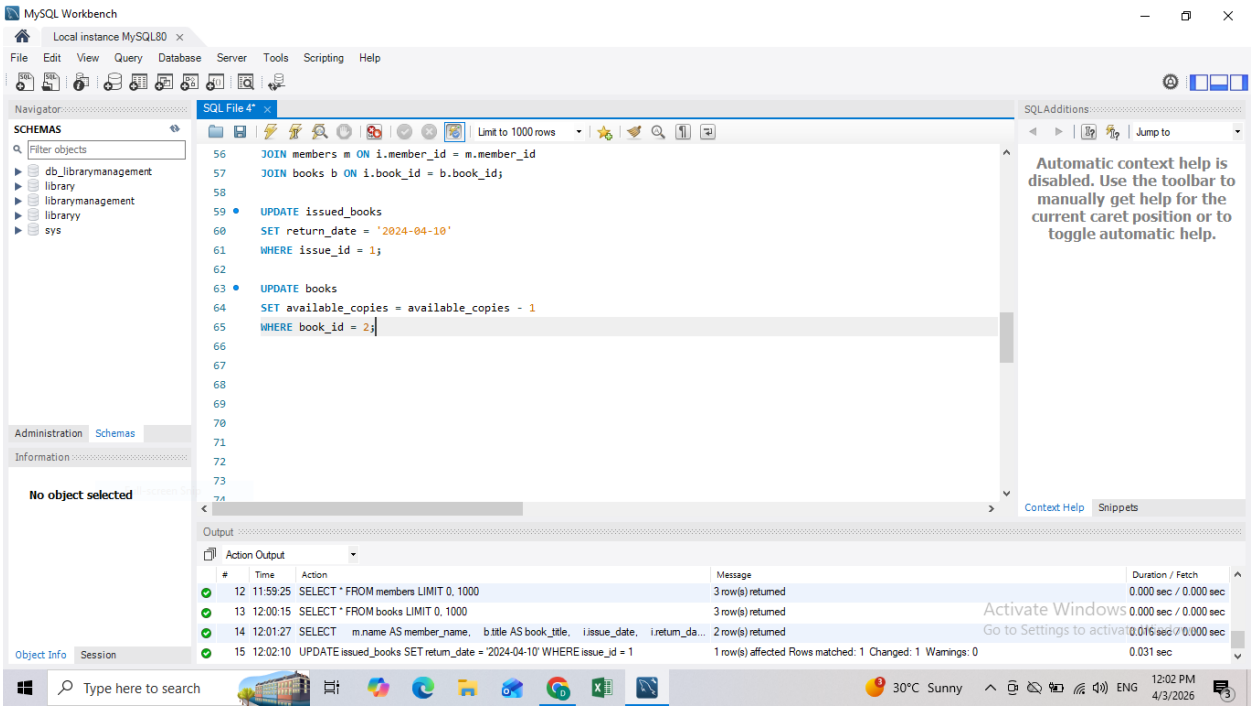




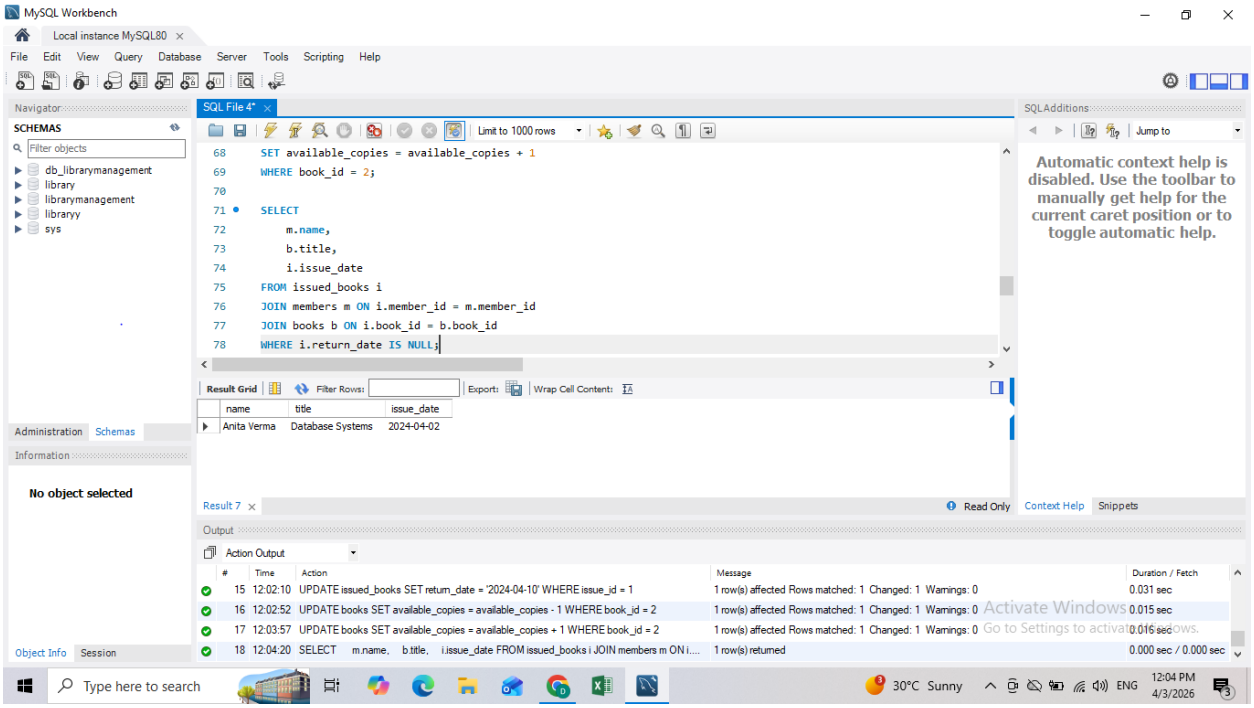
VIEWING ISSUED BOOKS WITH DETAILS



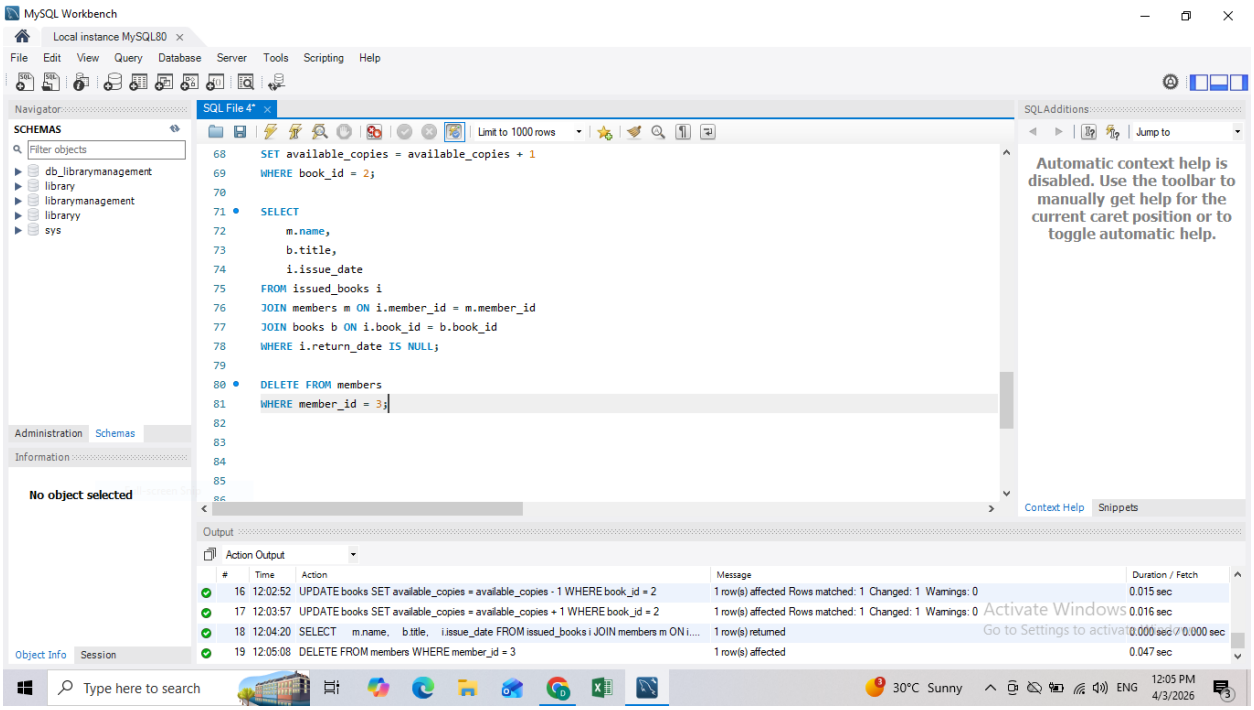
PERFORMING ACTIONS LIKE UPDATING RETURN DATE (BOOK RETURN), REDUCING AVAILABLE COPIES AFTER ISSUE, INCREASING COPIES AFTER RETURN



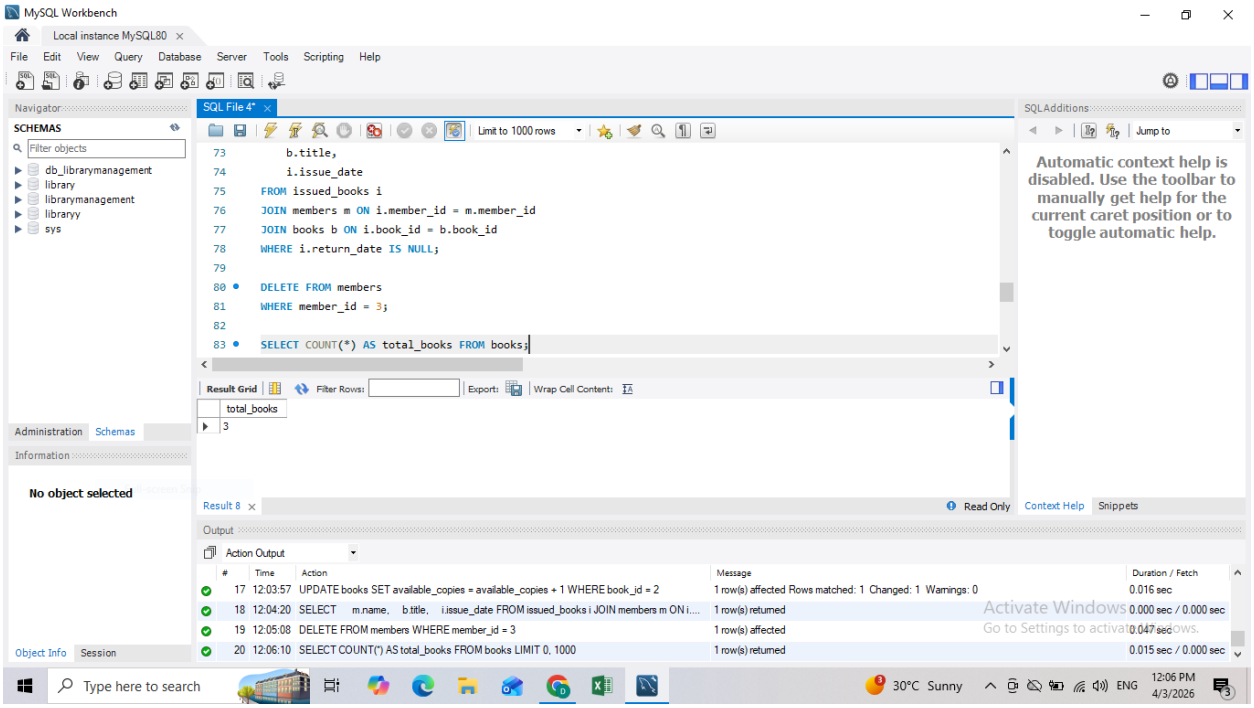
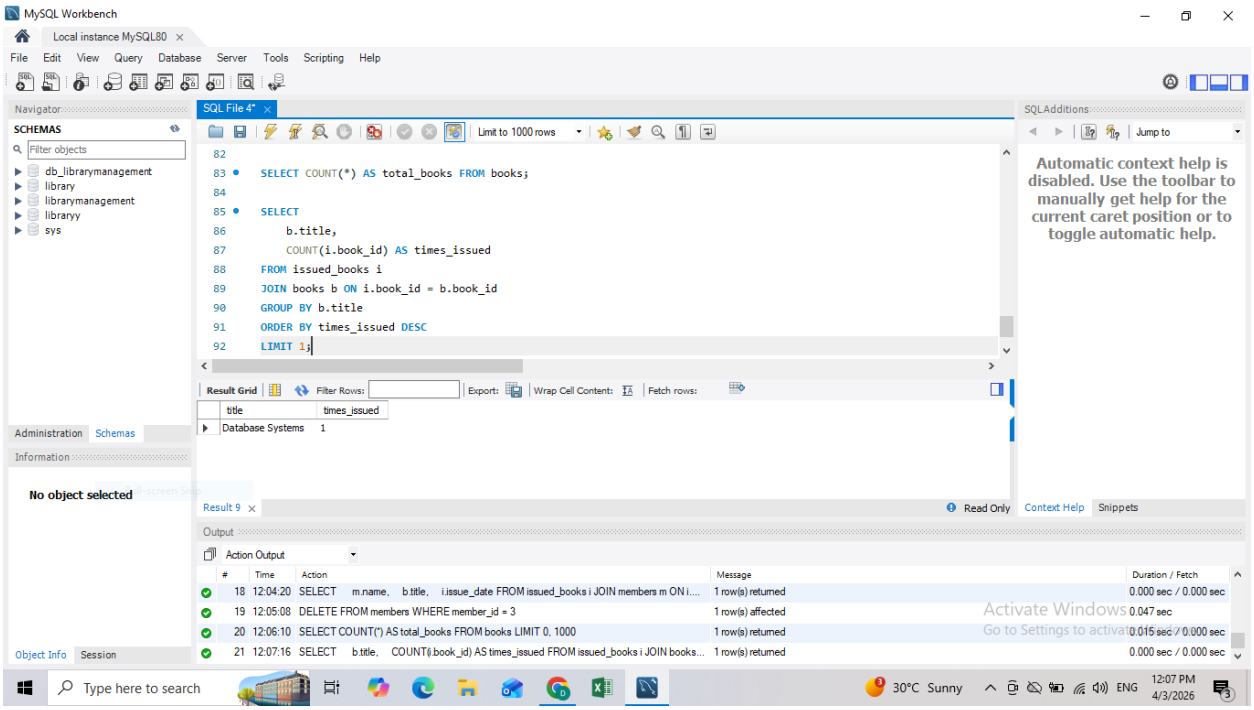
FINDING OVERDUE BOOKES (NOT RETURNED)



DELETING A MEMBER



COUNTING TOTAL BOOKS



FINDING MOST POPULAR BOOK